LESSON GUIDE

Technology Plus

Unit 7: Security

6.1.3 – Availability Concerns

Lesson Overview:

In this lesson, students will examine the Availability principle of the CIA Triad, focusing on how systems and data remain accessible to authorized users. Common threats to availability will be discussed such as natural disasters, hardware failures, DDoS attacks, and ransomware. Students also learn about mitigation strategies including backups, redundancy, cloud hosting, and more.

Students will:

- Define availability as it relates to the CIA Triad.
- Evaluate the impact of downtime on individuals or organizations.
- Identify common threats to availability.
- Examine best practices for ensuring availability.

Guiding Question: How do we ensure that systems and data are reliably available when users need them, and what happens when availability is compromised?

Suggested Grade Levels: 8-12

Technology Needed: Yes

Approximate Time Required: 60-65 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 6.1 Explain fundamental security concepts and frameworks.
    - Confidentiality, integrity, and **availability**

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Lesson Background

Background Information:

Availability is one of the three foundational principles of the CIA Triad with Confidentiality and Integrity being the other two. Ensuring continuous access to data and systems is critical, especially in situations where delays can have significant consequences, such as hospitals, financial institutions, and schools. Availability focuses on maintaining the system's uptime, minimizing interruptions, and recovering quickly from incidents. Threats to availability come in many forms, including hardware failures, power outages, and even natural disasters. Attacks like Distributed Denial of Service (DDoS) can disable systems and bring organization operations to a halt. DDoS attacks use multiple machines, sometimes upwards of several thousand to send requests to servers to overwhelm and ultimately crash them . Often the devices used to do this are compromised machines that have been programmed to act as an automated bot to perform the attack. Even IoT devices can be used to perform DDoS attacks. Ransomware is also a major threat to availability as it can be used to encrypt systems and devices, preventing users and employees from accessing data or systems. Organizations use a variety of tools and strategies to defend against these disruptions, such as backups, redundant systems, cloud services, and disaster recovery plans.

Materials Needed:

Materials per Class	• Lesson Guide 6.1.3 – Availability Concerns • Lesson Slides 6.1.3 – Availability Concerns • Lab Slides 6.1.3 – Denial of Service
Materials per Student	• Student Handout 6.1.3 – Availability Concerns • Student Notes 6.1.3 – Availability Concerns • Assessment 6.1.3 – Availability Concerns (Optional)

Lesson Background

Cyber Terms:

Term	Definition
Denial of Service (DoS)	an attack that floods a server or network with excessive traffic to overwhelm its resources and prevent it from being able to process legitimate traffic
Distributed Denial of Service (DDoS)	like a DoS attack, the end goal is to overwhelm a network or webserver, but the traffic is generated from multiple sources making it more difficult to mitigate
redundancy	inclusion of extra components, like backup servers or power supplies, that take over when the primary components fail
failover system	systems designed to automatically switch to a backup or secondary system in the event of a failure
load balancing	distributing network traffic across multiple servers to prevent overload and maintain uptime

6.1.3 – Availability Concerns

Guiding Question: How do we ensure that systems and data are reliably available when users need them, and what happens when availability is compromised?

Lesson Launch: (3 - 5 minutes)

- Introduce concerns with availability and why it is important to ensure systems and data are reliable.
- Have students discuss consequences if a service is down and in their experience why was it down, how long did it last, did it prevent them from completing any work, etc.
- Use the scenario to ask and discuss with students about their experience with attempting to access a website, network, or related system that failed to load or function.
- Have students discuss:
  - What type of site was it?
  - How frustrated did it make you?
  - What did it prevent you from doing?

Defending Against Downtime (20 - 25 minutes)

- Distribute Student Handout 6.1.3 – Availability Concerns Handout and encourage them to fill in answers as you go through today's lesson

Review of the CIA Triad and Availability

- Remind students that the CIA Triad focuses on confidentiality, keeping data secure from unauthorized viewers, integrity, keeping data trustworthy, and availability, keeping it accessible.
- Availability focuses on ensuring data and systems storing data is accessible to authorized users when needed. This includes internal and external users.
- Threats to availability can range from being accidental, controlled, or malicious in nature.

Availability Failures Consequences

- Availability failures often lead to organizational losses that could come in the form of:
  - Financial losses.
  - Reputational damage where trust is lost.
  - Operational disruption such as loss of resources or inability to perform day-to-day tasks.
  - Legal and compliance risks that could lead to fines or penalties.

Availability Concerns – Unexpected Issues

- Regardless of how robust a system is, unexpected issues can and will often appear.

- **Hardware issues** from devices crashing, overheating, or defects occur and can disrupt services.
- **Natural disasters** such as fires, flooding, earthquakes, etc. can damage resources and/or the structures housing resources.
- **Software issues** from misconfigurations, poor coding, or mismatched technical specifications can prevent access.
- **Power outages** can result from several sources leading to availability issues.

Availability Concerns – Controlled Issues

- **Maintenance** often occurs to perform updates or change equipment; however, it can lead to issues due to:
    - Extended amounts of time to complete the maintenance.
    - Unexpected problems when applying changes to the production side, or live side, that were not caught or experienced during testing.
    - Conflicts with other systems or services.

Availability Concerns – Malicious Activity

- **Malicious activity** aimed at availability can be some of the more detrimental attacks.
    - **Denial of Service (DoS)** or **Distributed Denial of Service (DDoS)** attack aim to overwhelm systems and/or networks with traffic or requests to the point that they can no longer handle processing legitimate traffic.
    - **Ransomware** encrypts files or systems preventing users from being able to use or access them.

Maintaining Availability – Fallback Systems

- **Backups** provide a copy of data should the data be lost or made unavailable. Backup conditions, such as how often it occurs, storage locations, and methods should be detailed in recovery plans.
- **Redundancy** means having a secondary system, such as the equipment used in a network, in place should the primary system be down.
- **Failover systems** activate automatically should a triggering action occur, such as an uninterruptible power supply (UPS) taking over if the primary power source goes down.

Maintaining Availability – Active Mitigation

- **Load balancing** spreads network traffic across multiple servers to prevent overloads or mitigate attacks.
- **Cloud services** allow storage of data offsite, with multiple redundancies and systems in place to handle issues threatening availability.

Ensuring Availability – Summary

- Compare the availability concerns, their descriptions, and real-life examples to their possible solutions and specific examples.
- Encourage students to provide additional solutions to the concerns.

Lab: 6.1.3 Denial of Service (25-30 minutes)

- This lab will use the Ubuntu VM and Apache webserver to host a webpage and the Kali Linux to perform an attack.

- Students will then launch a Denial of Service (DoS) attack against the Ubuntu machine leading to poor performance and delays in loading times for the webpage. The attack will use Slowloris, which connects to the webserver using broken web requests. These broken requests force the Apache webserver to keep the connections alive, consuming resources that could be used for legitimate requests.
- After seeing the effects of the attack, students will change the configuration of the Apache webserver and how it handles requests, specifically how long it keeps a connection alive before discarding it.
- Make sure students restart the apache2 server to apply the new configuration.
- With the new configurations, students will relaunch the attack and examine how the DoS attack is less effective.
- How did Slowloris perform a DoS attack on the Ubuntu/Apache webserver and what resulted from it?

    It sent packets to the server with broken headers, leading to the connection being maintained longer than it should, which in turn used up resources and slowed the server down.

- How did changing the timeout configuration help mitigate the DoS attack?

    It decreased the time the server would leave connections open, allowing those freed resources to handle legitimate connections. It improved the response time by about 50%.

- If you were running a secure server, using HTTPS, why would it be important to carefully test the timeout configuration values when trying to balance security and DoS (or DDoS) resilience?

    If the timeout configuration is set too low, there may not be enough time for the user's browser to check the certificate for the server to verify a secure connection. This could result in legitimate users having loading issues.

Putting It All Together (3-5 minutes)

Discussion Questions

- It's the last week of school and a ransomware attack hits the district, locking all grades, test scores, and other records. What could be done to reduce this threat aimed at availability?

    Possible mitigation could involve having backups to restore the data, i.e. grades, scores, and records, to their original state.

- What is a DDoS attack and how does it pose a risk to availability?

    Distributed Denial of Service aims to overwhelm a webserver or network with excessive traffic and prevent it from being able to process legitimate traffic and prevent users from accessing data or services.

- There is a local coffee shop that everyone loves, but they run everything for the business on one server. What are some availability concerns and how could you alleviate those concerns?

    Student answers will vary. They could have a hardware failure that disrupts all services or leads to a complete loss. Having a secondary system could help by taking over should the primary go down. An attack could prevent services from occurring. Using cloud services could alleviate this concern.